

WEB DEVELOPMENT • SESSION 02

Organizing Content & Text Superpowers

Mastering Block vs Inline Elements, Advanced Lists, Formatting & Entities

Instructor Edition | Hands-On Mentorship Track

How Browsers See Content: Block vs. Inline

■ Block Elements

Takes 100% full screen width available.

Always forces a fresh new line break above and below.

Next elements stack below, never side-by-side.

Examples: `<h1>` to `<h6>`, `<p>`, `<div>`, `<ul>`, `<ol>`, `<li>`, `<hr>`

🔍 Real-World Analogy: Solid bricks or shipping containers stacked on top of each other.

● Inline Elements

Takes only the exact width of its own content.

Sits side-by-side with other inline tags on the same line.

Never starts a new line automatically.

Examples: `<a>`, `<img>`, `<strong>`, `<em>`, `<span>`, `<mark>`

🔍 Real-World Analogy: A highlighter pen stroke or inline label stickers inside a paragraph.

Styling Meaning Into Words

`<strong>` vs `<b>`

Visual Bold Effect

`<b>` only renders visual thickness. `<strong>` signals urgent structural importance to screen readers and SEO engines.

`<em>` vs `<i>`

Visual Italic Effect

`<i>` simply slants text. `<em>` conveys verbal vocal stress and contextual emphasis in conversation.

`<mark>`

Highlighter Effect

Applies an authentic fluorescent yellow highlighter stroke across targeted phrases without needing CSS.

`<del>`

Strikethrough Effect

Draws a strikethrough line over deleted/outdated data (perfect for sale discounts: Rs. 5000 -> Rs. 3500).

Lists in HTML: Unordered vs. Ordered

Unordered List (`<ul>`)

- Used when order / sequence does NOT matter.
- Examples: Shopping lists, ingredients, navigation bar links.
- Items are wrapped in `<li>...</li>` tags.

Code Example:

```
<ul>
  <li>Melted Butter</li>
  <li>Brown Sugar</li>
  <li>Chocolate Chips</li>
</ul>
```

Ordered List (`<ol>`)

- Used when sequential priority is mandatory.
- Examples: Recipe steps, tutorials, countdown rankings.
- Key Attributes: `type="A"` (letters), `start="5"` (custom start).

Code Example:

```
<ol type="1">
  <li>Preheat oven to 350°F</li>
  <li>Mix butter and sugar</li>
  <li>Bake for 12 minutes</li>
</ol>
```

Advanced Lists: Nested & Description Lists

Nested Lists (Sub-Menus)

★ **Golden Rule:** Always nest the sub-list **INSIDE** the parent `<li>` before closing it!

```
<ul>
  <li>Frontend
    <ol>
      <li>HTML5</li>
      <li>CSS3</li>
    </ol>
  </li>
  <li>Backend</li>
</ul>
```

Description Lists (`<dl>`)

Perfect for glossaries, dictionaries, and key-value pairs:

- `<dl>` : Description list container
- `<dt>` : Description Term (The word/label)
- `<dd>` : Description Data (The definition)

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Anchors & Images: The Connected Web

Anchor Tag (<a>)

```
<a href="https://google.com"
  target="_blank">
  Visit Google
</a>
```

- href (Hypertext Reference): The destination URL link.
- target="_blank": Opens the link in a fresh new tab so the student/user does not leave your website.
- Inline element: Sits smoothly inside any sentence or paragraph.

Image Tag ()

```

```

- Void Tag: Self-closing, never has an tag.
- src (Source): Path to file (local image or web URL).
- alt (Alternative Text): MANDATORY for accessibility, screen readers for visually impaired, and broken images fallback.

HTML Entities & Code Comments

HTML Entities (Special Symbols)

Used when characters clash with HTML code brackets:

<code>&lt;</code>	→	<code><</code>	(Less than symbol)
<code>&gt;</code>	→	<code>></code>	(Greater than symbol)
<code>&amp;</code>	→	<code>&</code>	(Ampersand symbol)
<code>&copy;</code>	→	<code>©</code>	(Copyright symbol)
<code>&nbsp;</code>	→		(Non-breaking extra whitespace)

Code Comments (`<!-- -->`)

- Completely invisible on the website screen.
- Only visible to developers reading the source code.
- Used to document sections, explain logic, or temporarily disable code.

```
<!-- Main Projects Section Starts -->
```

```
<section>
```

```
  <h2>My Work</h2>
```

```
</section>
```

```
<!-- End of Projects Section -->
```

⚡ **VS Code Shortcut:** `Ctrl + /` (Win) or `Cmd + /` (Mac)

Live Challenge

📋 In-Class Task: Frontend Tech Glossary

1. Main title wrapped in an `<h1>`.
2. Description List (`<dl>`) defining HTML, CSS, & JS.
3. Nested Unordered List outlining 3 coding goals.
4. External link to freeCodeCamp with `target="_blank"`.
5. Footer paragraph with a copyright entity (`©` 2026).

🏆 Weekly Assignment 1

Submission Deadline Before Next Class